

Mutual Evaluation: Measuring Agent Reliability Without Ground-Truth

Zachary Robertson
April, 2026

Domains without Reliable Ground-Truth

Current Situation

- Researchers increasingly use AI agents to evaluate agents
- In many domains, evaluator judgements are not reliable
- Variance across evaluators is often summarized, not studied

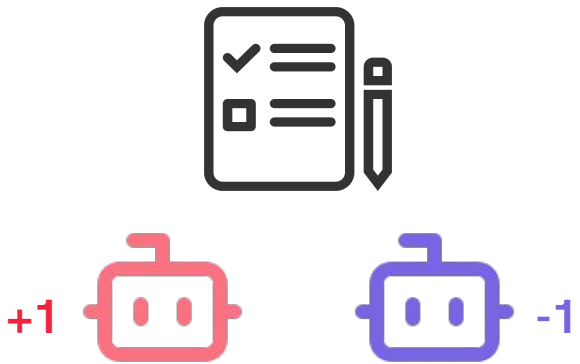
The Core Problem: Evaluation Depends on the Evaluator

- An agent score reflects both:
  - The system being evaluated
  - The evaluator applying the metric
- If evaluators disagree, that is evidence about the evaluation process itself
- We need a way to measure reliability **without assuming a gold evaluator**

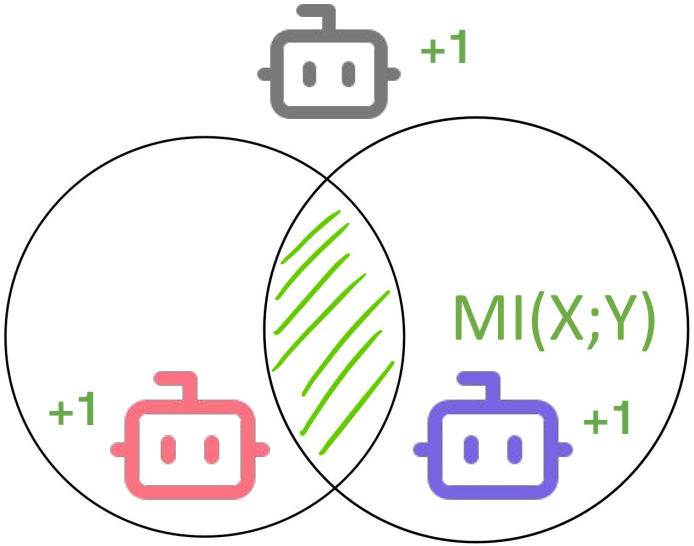
Is There Another Way? Preference vs. Mutual Evaluation

Instead of asking “which output is better?”—a question that can be gamed—we ask: **can these outputs plausibly come from the same source?**

Preference Evaluation (Zero-Sum)



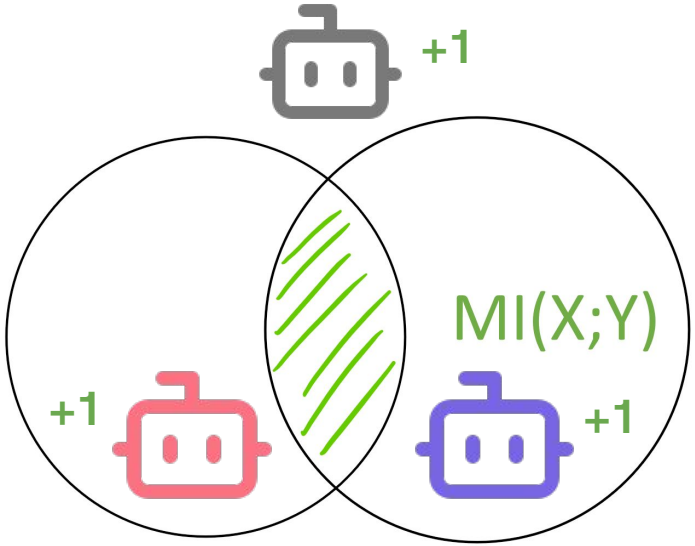
Mutual Evaluation (Cooperative)



This Talk

1. Why Mutual Evaluation?
2. Mutual Evaluation as a Protocol
3. Three Application Templates

Why Mutual Evaluation?



1. Measures agents **and** evaluator
2. Treats disagreement as evidence, not just noise
3. Does not require trusted ground truth labels

Why Mutual Evaluation? - A Shift in What We Measure

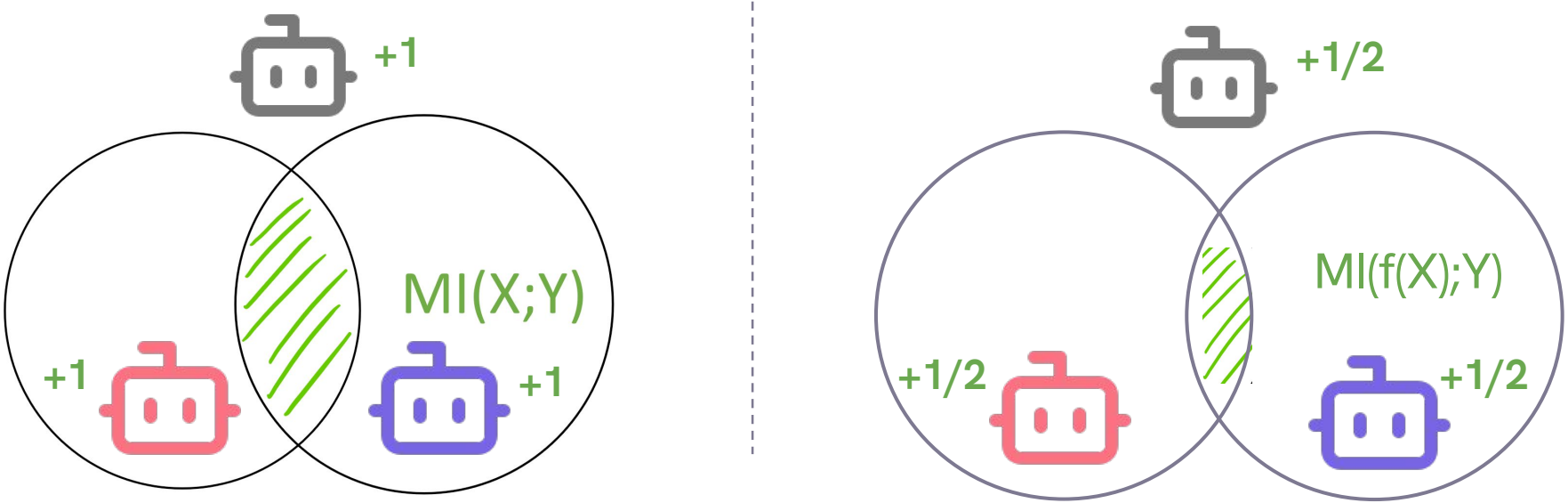
Standard view

- Evaluator judgement is treated as gold standard
- Agent quality is inferred from evaluator judgement

Mutual evaluation

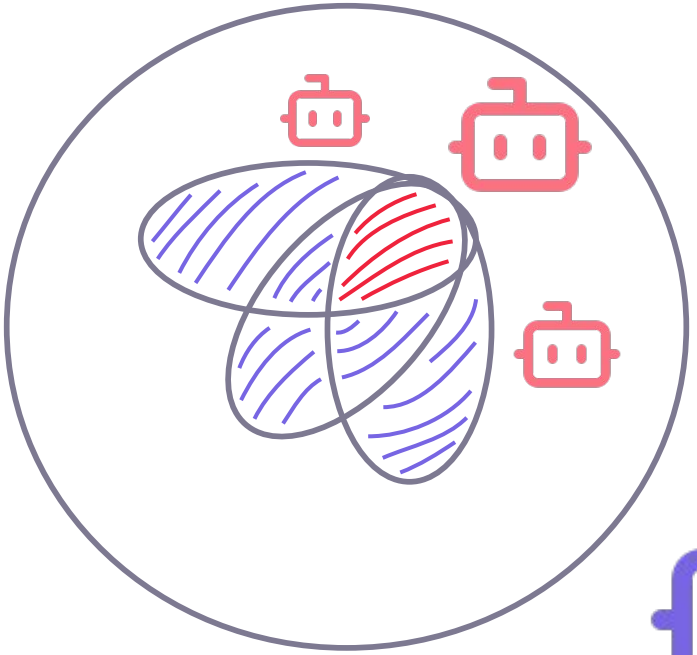
- Multiple agents produce outputs
- Multiple evaluators apply a common protocol
- Agreement and disagreement patterns become object of study

Why Mutual Evaluation? - A No Post-Processing Incentive



If the critic is accurate, agents don't gain by removing information

Mutual Evaluation Does Not Necessarily Reward Majority



A conclusion can be common but uninformative

A conclusion can be less common but highly diagnostic



Rewards preserved structure **across** outputs

Specification Driven Implementation

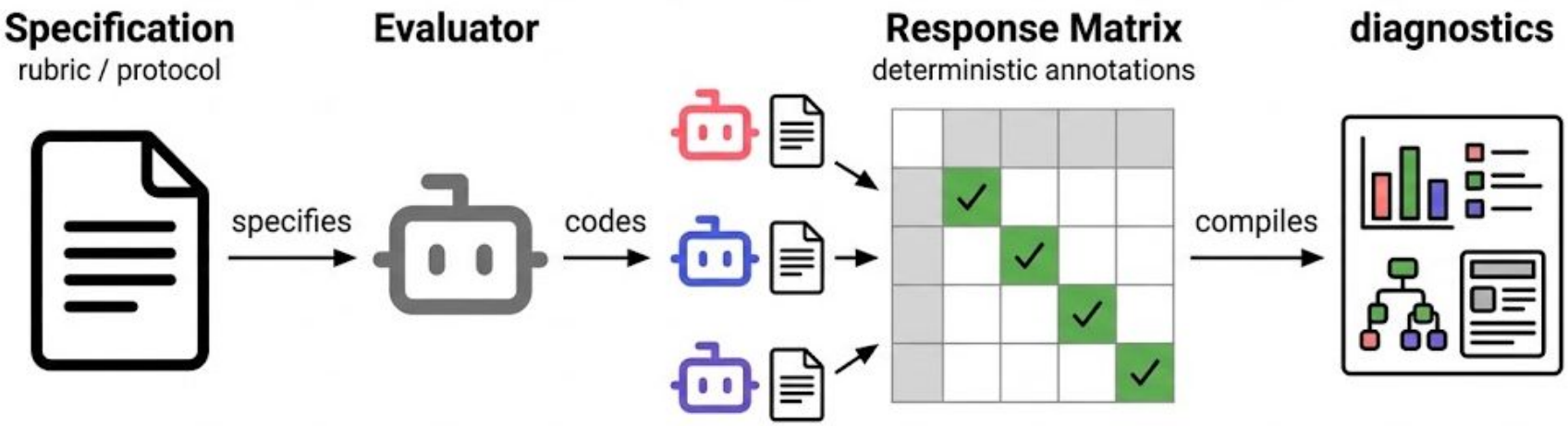
Implementation

1. Agents produce natural-language outputs
2. Evaluators create rubrics / coding schemes for deterministic annotations
3. The annotations define a response matrix
4. Aggregation analyzes the structure induced by those annotations

Spec-Driven Implementation (Protocol)

Evaluator Type Coding

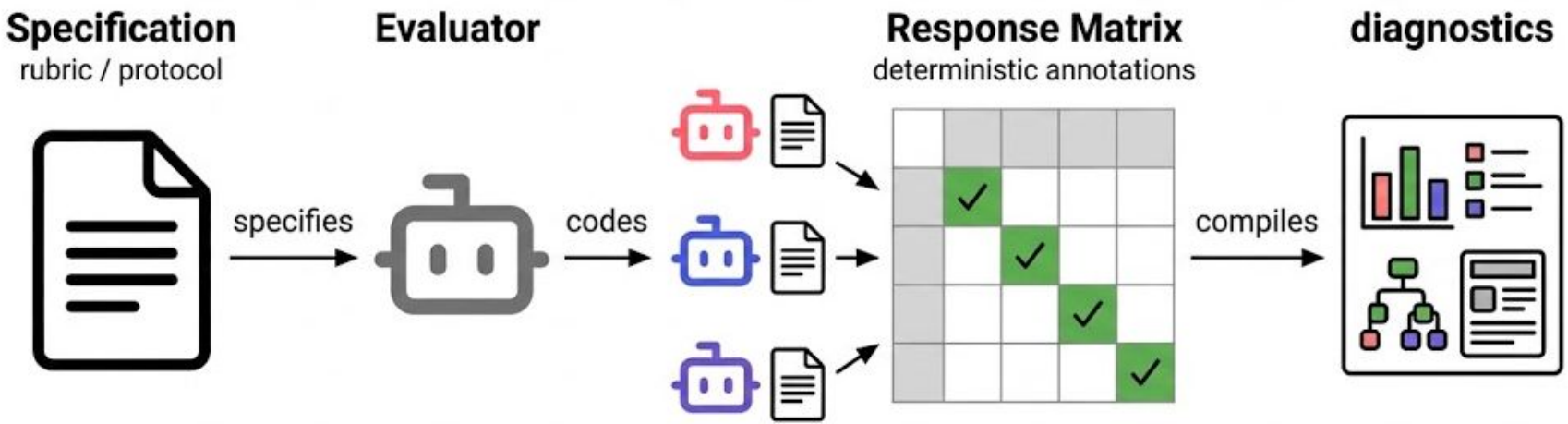
Reliability Analysis



Spec-Driven Implementation (Protocol)

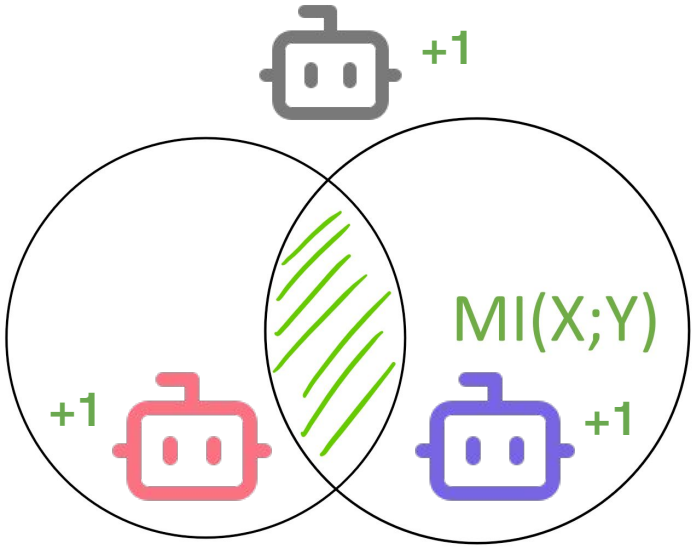
Evaluator Type Coding

Reliability Analysis



What construct are we measuring?

Mutual Evaluation (Cooperative Game)



Definition: The **evaluation-welfare** of a mutual evaluation game is the advantage of an optimal classifier in distinguishing genuine agent response pairs from pairs formed by independently sampling agent responses from different items.

Three Application Templates

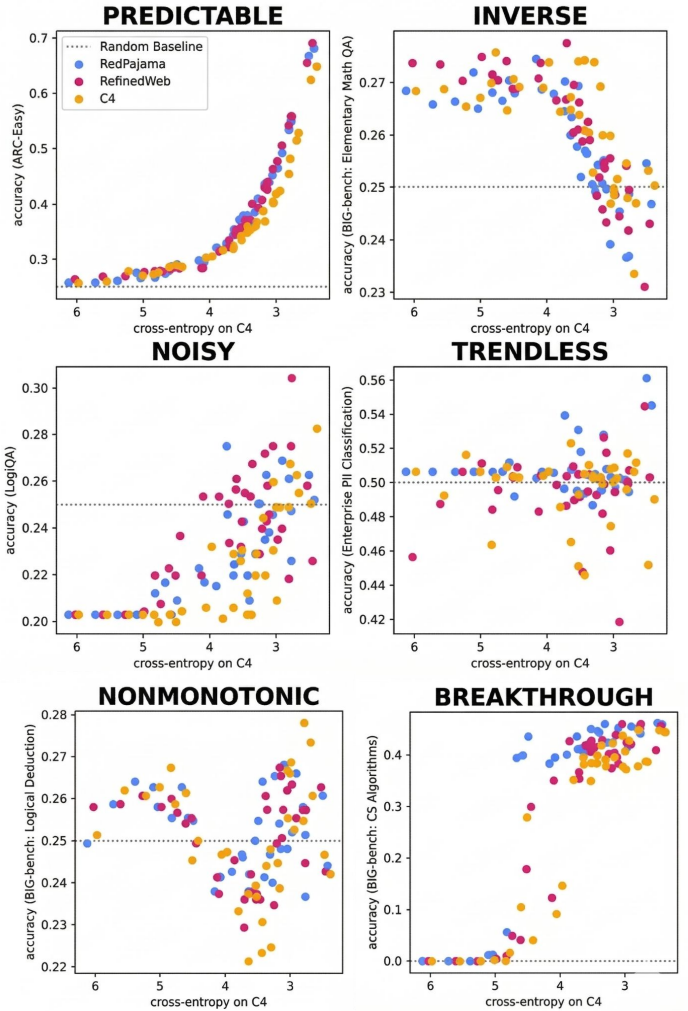
- **Finding reproducibility**
  - Do independent agents with the same data reproduce the same finding?
- **Specification robustness**
  - Does disagreement reveal hidden methodological ambiguity in the specification?
- **Evaluator replicability**
  - Do independent evaluators looking at many reproduced findings replicate similar interpretations?

Experiment Design

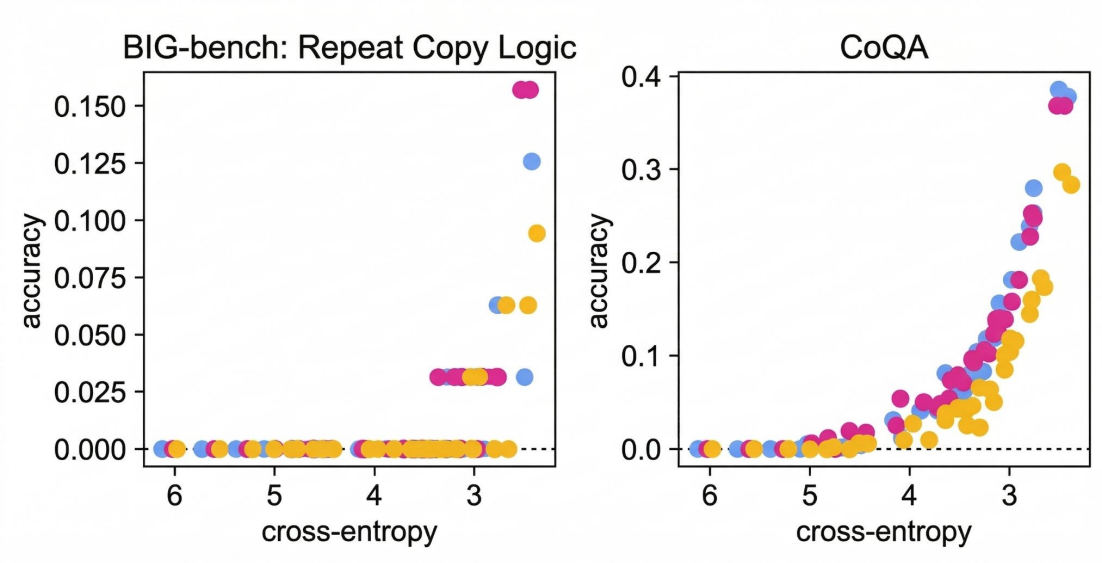
- **Domain Selection:**
  - Scaling-law classification
  - TMLR editorial timeline audit
  - BenchPress benchmark prediction
- **Agent Taxonomy:**
  - 1 human rule + 4 GPT-o3 annotators
  - 36 independent Claude analysis agents + 1 evaluator
  - 48 analysis agents + 2 independent evaluators
- **Evaluation Metrics and Comparisons:**
  - Pairwise reliability / mean annotator reliability (welfare)
  - TVD-MI to implement reliability metric / permutation null
  - Cross-evaluator correlation

Template 1: Finding Reproducibility

- Lourie, Hu, and Cho (EMNLP Findings 2025) classify downstream LLM scaling plots and conclude only 39% are “predictable”
- But “predictable after transformation” is described verbally and judged visually
- **Question:** can independent annotators reproduce the same analysis rule?

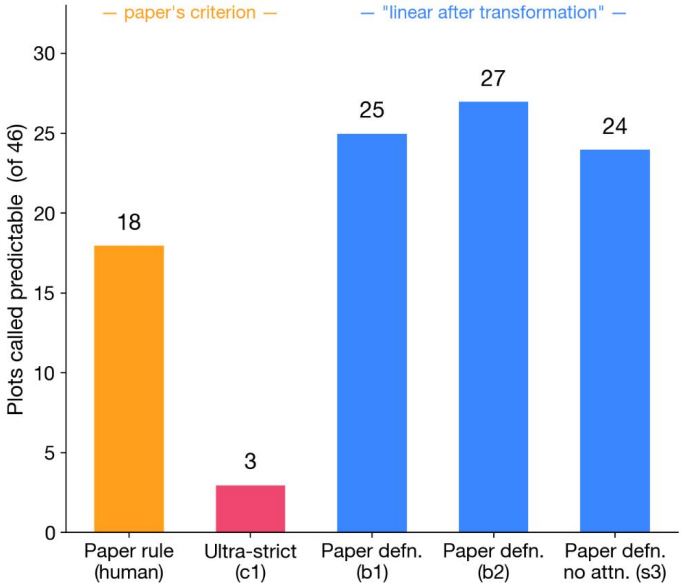


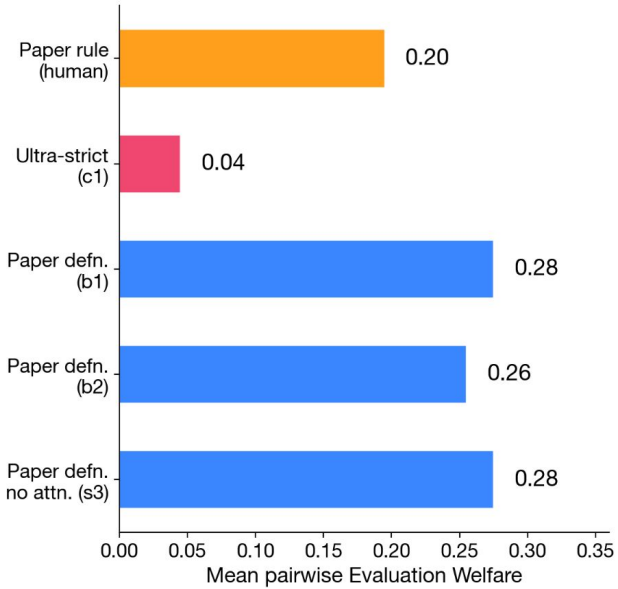
The 39% headline is not measurement-stable



- The authors label both plots above as “predictable”
- AI labels only CoQA as “predictable”

The 39% headline is not measurement-stable





- Counts of “predictable” swing from 3 to 27 under reasonable interpretations
- The paper’s grouping is not reproducible enough to function as ground truth

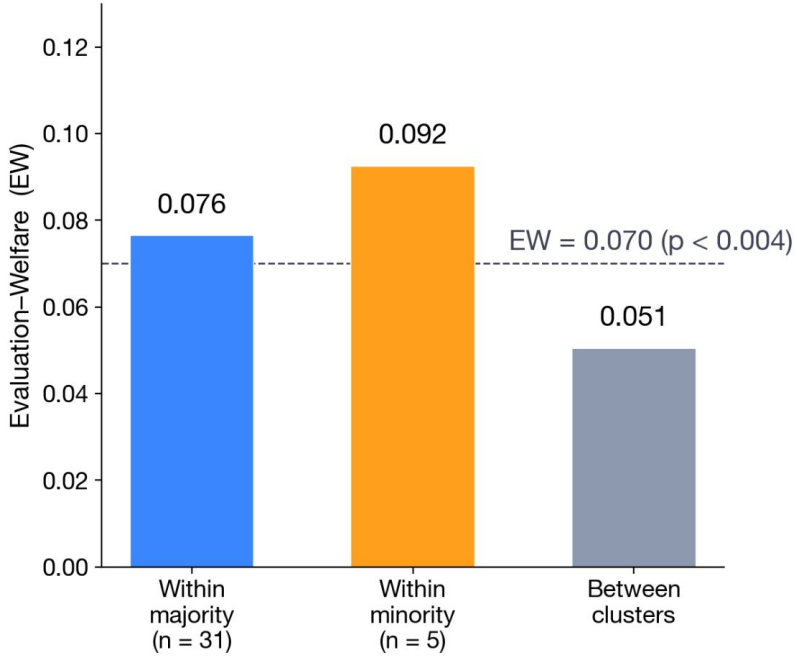
Template 2: Specification Robustness

- TMLR's action editor guidelines state 5 weeks between reviews and submitted decision
- **Question:** is this specification sensitive to phrasing?

Based on their discussion with the authors, reviewers will submit their own recommendation to you on a decision for a submission no sooner than 2 weeks and no later than 4 weeks after the submission of the third review.

Once all 3 reviewer decision recommendations have been submitted, you should aim to submit your decision proposal within 1 week. Your decision proposal should therefore be submitted within 5 weeks of the beginning of the discussion between authors and reviewers.

Agent disagreement reveals a specification ambiguity



Majority cluster

n = 31 (86%)

```
3rd review → decision
Median  45.3 days
> 35 d  82.5%
```

Minority cluster

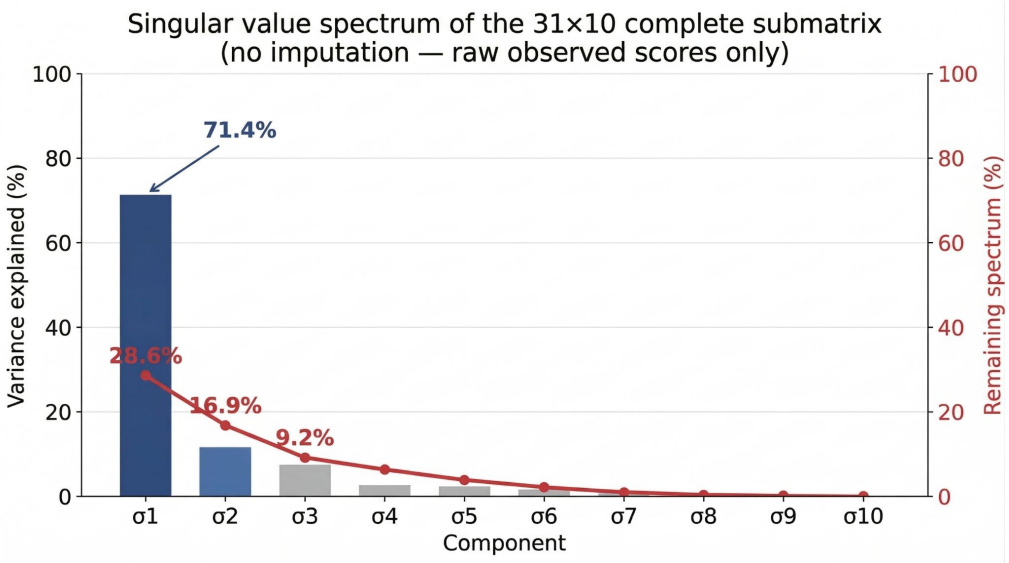
n = 5 (14%)

```
1st review → decision
Median  69.0 days
> 35 d  95.7%
```

- The minority scores **higher** than majority
- Clear split around the interpretation of when the “clock” starts

Template 3: Evaluator Replicability

- BenchPress starts from a plausible object-level claim: “you don’t need to run every eval”
- The observed 31x10 matrix has >90% variance explained with three singular components
- **Question:** will evaluators of agents reproducing “low-rank” findings replicate this interpretation?

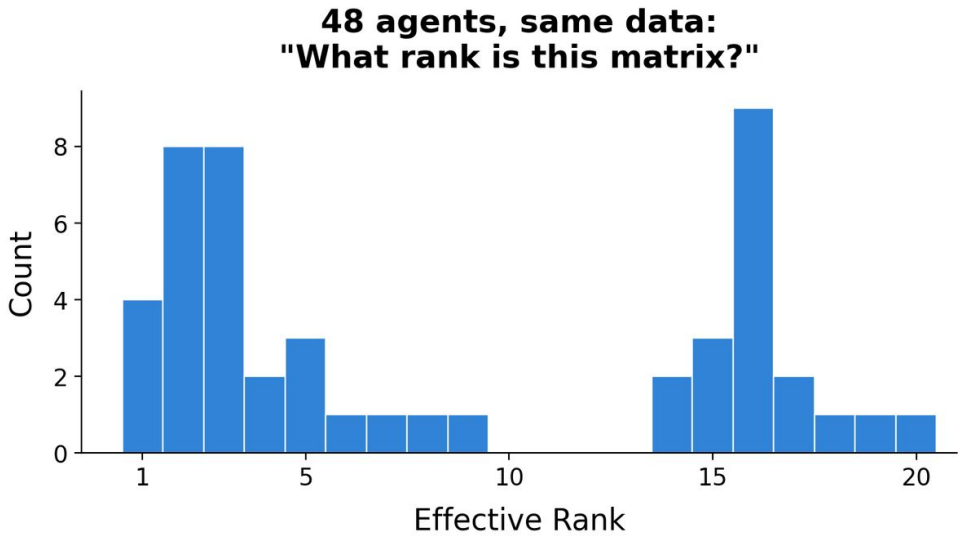


Our pre registered replicability hypotheses

1. At least 75% of agents report effective rank ≤ 3.
2. The benchmark subset selected replicates (mean Jaccard > 0.2)
3. Showing 5 benchmarks per LLM yields MAE < 10 with probability at least 80%
4. Preprocessing is identified as a primary fork
5. At least 90% of agents support the claim the matrix is low-rank and benchmark performance is predictably structured

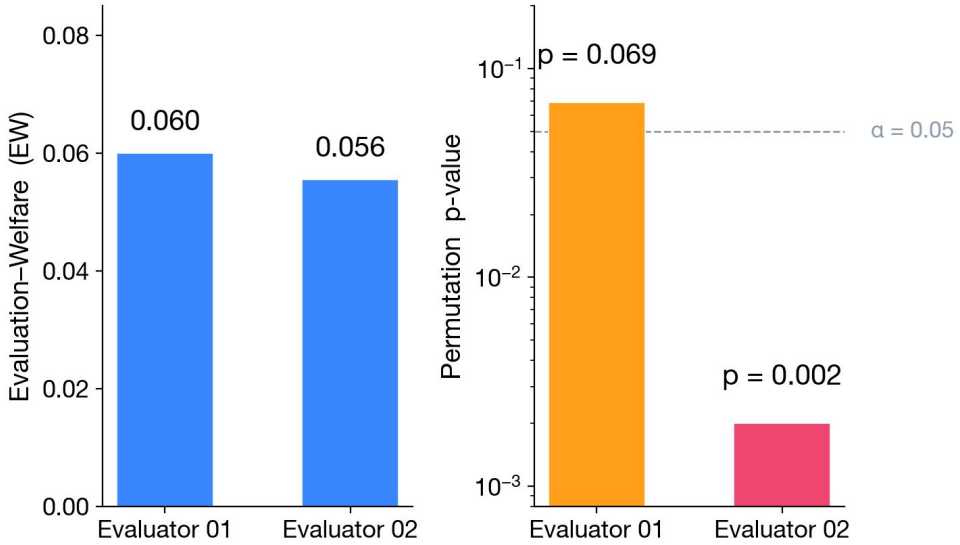
The low-rank claim is not evaluator independent

- All five claims we hypothesize replicate fail
- Only 42% of agents report effective rank ≤3
- Subset selection has Jaccard overlap ~0.12
- **Conclusion:** the interpretation needs nuance



The low-rank claim is not evaluator independent

- Independent evaluators correlate only modestly (r = 0.6)
- Evaluator qualitative coding can score agent welfare similarly, but diverge in significance
- The evaluator side interpretations are not evaluator independent

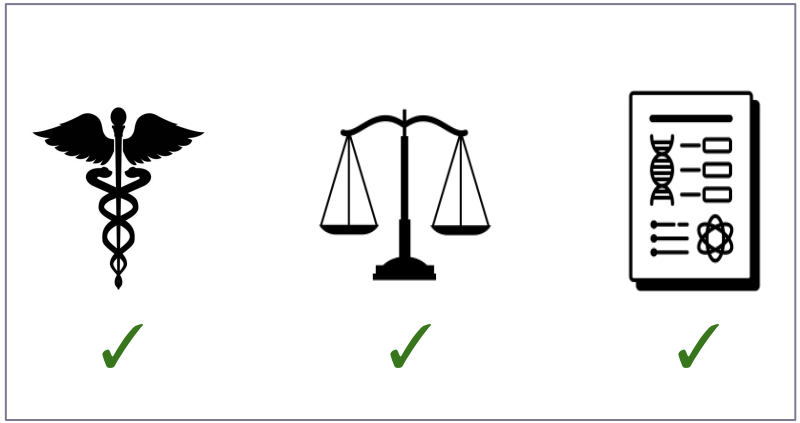


Outlook: Nonstandard Errors in Agent-Evaluator Systems

- Agent-evaluator systems given identical data and instructions can disagree substantially
- Measuring dispersion alone is not enough
- We propose **NSE-Bench** to measure:
  - How much disagreement?
  - Does disagreement have structure?
  - Which methodological forks carry signal?

Conclusions

1. **Mutual evaluation** is a way to measure the reliability of the evaluation system itself
2. **Highlights internal validity** as a limiting factor
3. **NSE-Bench** is a step toward making agent (dis)agreement measurable for domains without ground truth





Conclusion

Thank You

Stanford

(Paper)

(Code)

- **Contact:** zroberts@stanford.edu
- **ArXiv:** “Let’s Measure Information Step-by-Step: LLM-Based Evaluation Beyond Vibes” - https://www.arxiv.org/abs/2508.05469
- **NSE-Bench** - https://github.com/zrobertson466920/nse-bench
- **Collaborators:** Sanmi Koyejo, Hansol Lee, Suhana Bedi, Andrew Seha, Hannah Sha